

Calculated Field

Calculations which are used to find certain information that is not available in the data



Use of Calculated Fields



Calculated Field helps to create new data from the data that already exists in the data source.

When a calculated field is generated, basically a new field (or column) is created in the data source. The values or members of this calculated field are determined by the calculation that user controls.

This new calculated field is saved to the data source and can be used to create more robust visualizations.

But don't worry, the original data remains untouched.

Types of Calculated Field

Numerical
Field



Date
Field



Aggregate
Field



String
Field



Logical
Field



Dataset : India Life Expectancy

Life expectancy is a statistical measure of the average time a person is expected to live, based on the year of its birth, its current age, and other demographic factors including gender.

The dataset available on [Kaggle](#) has been modified for this chapter. List of columns used are as follows:-

- ❖ City Name
- ❖ State
- ❖ Last Election Date
- ❖ Death Rate
- ❖ Population Current



NUMERICAL FIELD



Steps to create Numerical Field

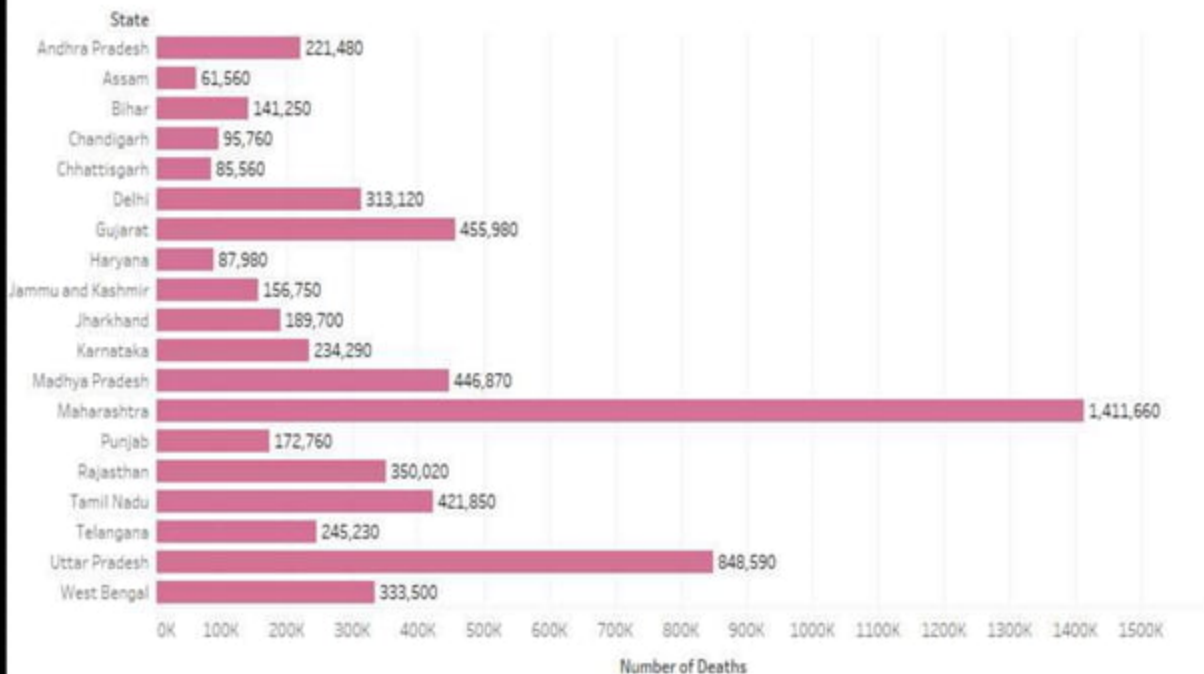
- Click the drop down beside the 'Dimensions' section.
- Select 'Create Calculated Field...' and a dialogue box will appear. Name it 'Number of Deaths' and write the formula:
$$([Death\ Rate]/100)*[Population\ Current]$$
-> Apply -> Ok.

This will create a field under 'Measures' section which describes the total number of deaths in every city.

- Create a Bar graph using 'State' and 'Number of Deaths'.

Resulting graph has been put up on the next slide and is self explanatory.

Numerical Field



Sum of Number of Deaths for each State. The marks are labeled by sum of Number of Deaths.



DATE FIELD



Steps to create Date Field

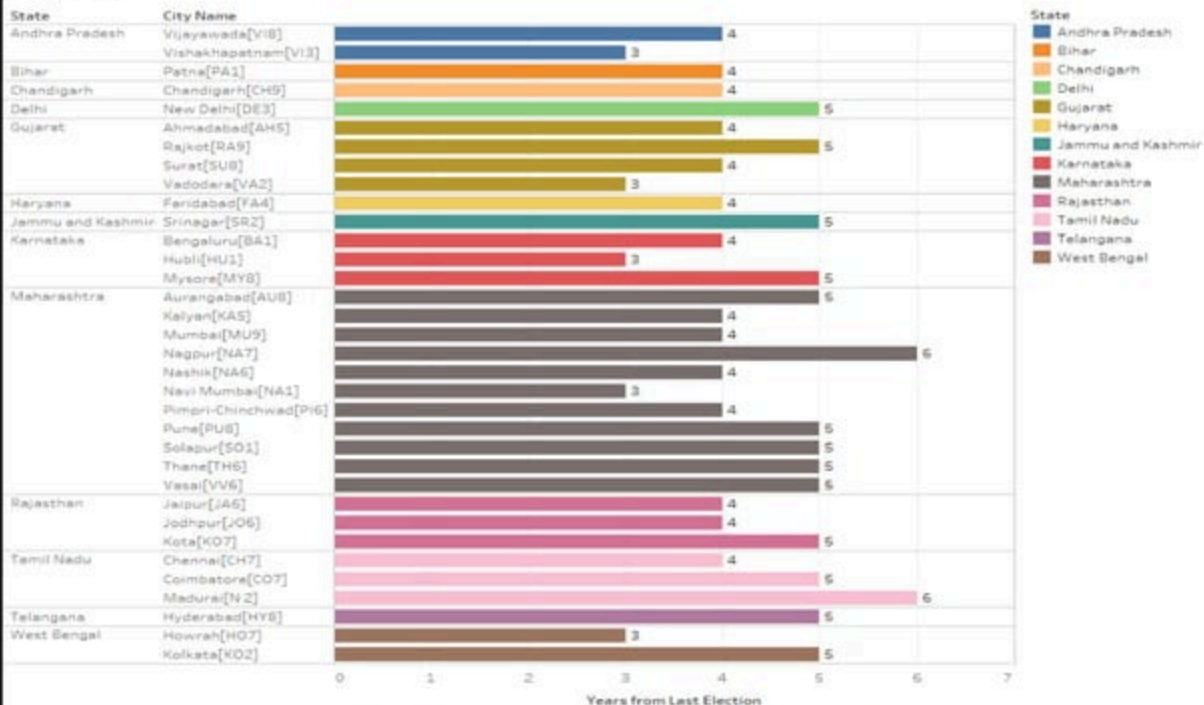
- Select the drop down beside 'Dimensions' section.
- Select 'Create Calculated Field...' and a dialogue box will appear. Name it 'Years from Last Election' and write formula: DATEDIFF('year' , [Last Election Date] , Today()) -> Apply -> Ok.

This will create a field under 'Measures' section which describes number of years that have passed since the last election was held in that City .

- Create a Bar graph using 'State', 'City' and 'Years from Last Election'.

Resulting graph (excluding few States) has been put up on the next slide for reference and is self explanatory.

Date Field



Sum of Years from Last Election for each City Name broken down by State. Color shows details about State. The marks are labeled by sum of Years from Last Election. The view is filtered on State, which keeps 13 of 19 members.



AGGREGATE FIELD



Steps to create Aggregate Field

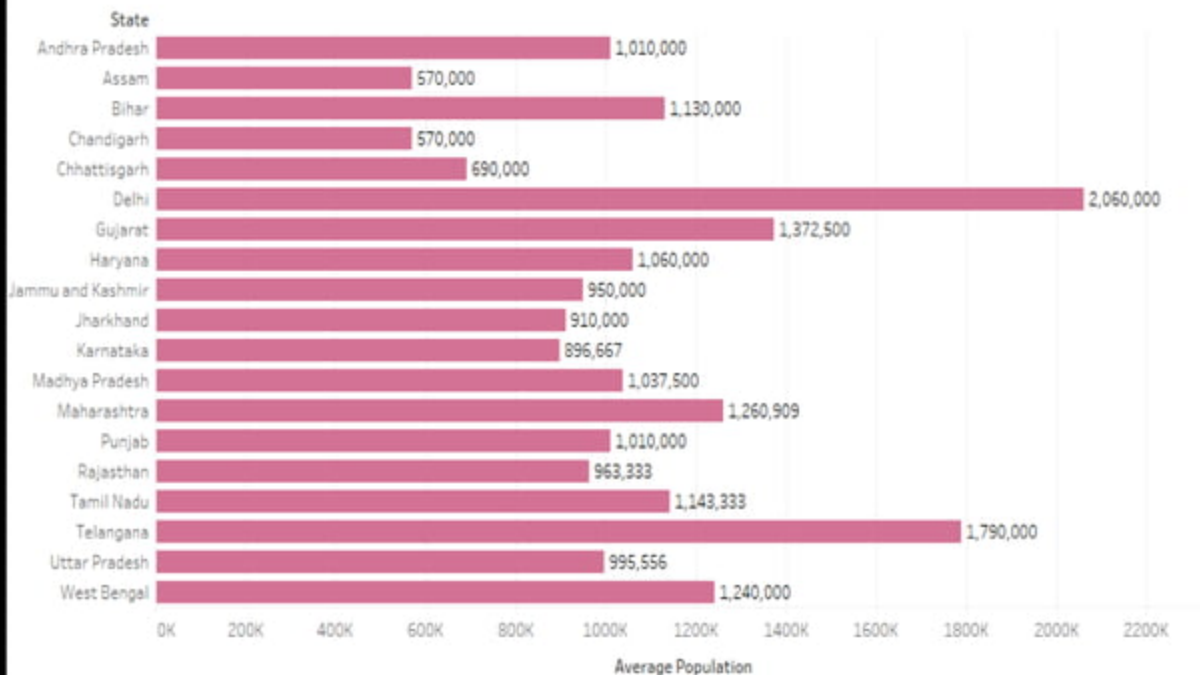
- Select the drop down beside 'Dimensions' section.
- Select 'Create Calculated Field...' and a dialogue box will appear. Name it 'Average Population' and write formula:
AVG ([Population Current]) -> Apply -> Ok.

This will create a field under 'Measures' section which describes the average population for every City.

- Create a Bar graph using 'State' and 'Average Population'.

Resulting graph has been put up on the next slide for reference and is self explanatory.

Aggregate Field



Average Population for each State. The marks are labeled by Average Population.



STRING FIELD



Steps to create String Field

There are two ways to create a String field in 'Dimensions' section.

One way is by Splitting an existing field and the other way is by Combining any two fields.

Steps to create a String field by splitting :-

- Right click on 'City Name' in 'Dimensions' section.
- Select 'Transform' and then 'Split'.

This will create two fields under 'Dimensions' section, one containing the name of the city and the other its code.

- Rename the 'City Name - Split 1' as 'City' and 'City Name - Split 2' as 'City Code'.

Steps to create String Field

Steps to create a String field by combining :-

- Select 'State' and 'City Name' using Ctrl key.
- Right click on either of the field. Select 'Create' and then 'Combined Field'.

This will create a field under 'Dimensions' section which combines the 'State' and 'City Name' using ',' as a separator for each row in the dataset.

- Create a Bar graph using 'City', 'City Code', 'City Name & State (Combined)' and 'Population Current'.

Resulting graph (excluding few States) has been put up on the next slide for reference and is self explanatory.

String Field



Sum of Population Current for each City Name & State (Combined) broken down by City and City Code. Color shows details about State. The marks are labeled by sum of Population Current. The view is filtered on State, which excludes Gujarat, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh.



LOGICAL FIELD



Steps to create Logical Field

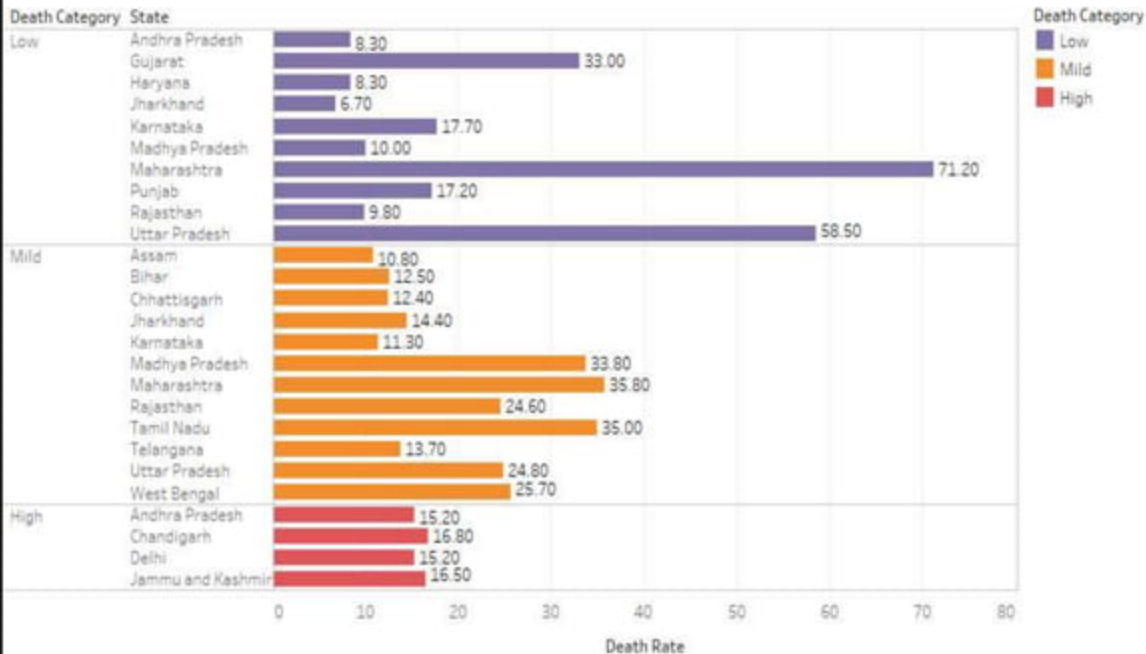
- Select the drop down beside 'Dimensions' section.
- Select 'Create Calculated Field...' and a dialogue box will appear. Name it 'Death Category' and write formula:
IF [Death Rate] <= 10 then 'Low'
ELSEIF [Death Rate] > 10 and [Death Rate] <= 15 then 'Mild'
ELSE 'High' END
- Click 'Apply' and then click 'Ok'.

This will create a field under 'Measures' section which labels the each row according to the above logic 'Death Rate'.

- Create a Bar graph using 'State', 'Death Rate' and 'Death Category'.

Resulting graph has been put up on the next slide for reference and is self explanatory.

Logical Field



Sum of Death Rate for each State broken down by Death Category. Color shows details about Death Category. The marks are labeled by sum of Death Rate.